
IC 150: Computation for Engineers

Course Code	IC 150
Course Name	Computation for Engineers
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum
Discontinued	17th Senate (Aug 2018)

Course Contents

- **Computers, programming and environment:** Computer and its components, common uses of a computer, computer as a machine, what is a program, program testing and verification, problem solving and implementation of algorithms, limitations of computing with computers, compilers, operating system/unix environment, editors, IDE's [3-4 Lectures]
- **Programming:** Problem solving with programming, Basics of Programming, Primitive types, Expressions, Decision making, Iteration, Function, Recursion, Pointer, Array, Structure & Union, Basic I/O, File handling, Dynamic Memory Allocation. [20-22 Lectures]
- **Numerical Computation and number crunching (Scilab/Python and Openoffice/Excel):** Scilab fundamentals, programming with Scilab, error handling, finding roots (various methods), matrix operations, Entering and Formatting Data and formulae, Using Built-in Functions, performing logical tests, interpolations [10-12 Lectures]
- **Reporting of Results:** Units, Significant figures, Graphs and tables for data presentation [1-2 Lectures]

Books and References

1. V. Rajaraman, **Computer Programming in C**
2. R. G. Dromey, **How to Solve It By Computer**
3. Kernighan and Ritchie, **The C Programming Language**
4. Kernighan and Pike, **The Unix Programming Environment**
5. Joseph C. Musto, William E. Howard, Richard R. Williams, **Engineering Computations: An Introduction Using MatLAB and Excel**, Tata McGraw Hill.